Dunman Secondary School
Secondary 4
G3 Additional Mathematics

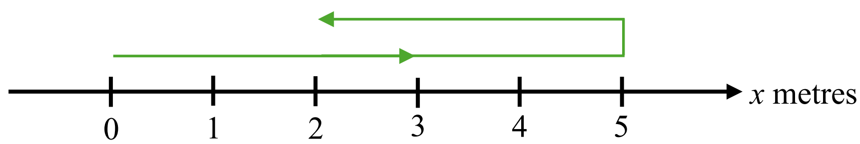
Chapter 16: Kinematics

Name: ______________________________ (   )   Class: ____________

16.1 Key Concepts in Kinematics

(A) Displacement

A boy runs along a straight path and make a U-turn at the 5th metre. His journey is recorded below:



1) Distance travelled =

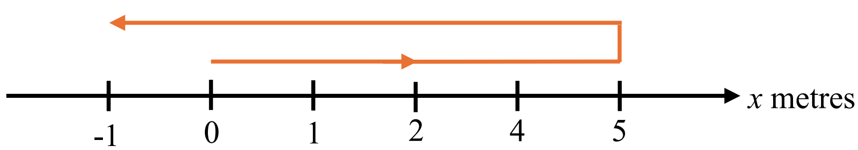
The displacement of the boy is the difference between the final position and the original position.

It is a vector quantity because it has both magnitude and direction.

2) Displacement, s =

Note: We often use "s" to denote displacement

The following shows another boy's running journey.



3) Displacement, s =

Note:

- If $s < 0$, the body is along the negative s-axis
- If $s = 0$, the particle is at the fixed point O
- If $s > 0$ the body is along the positive s-axis

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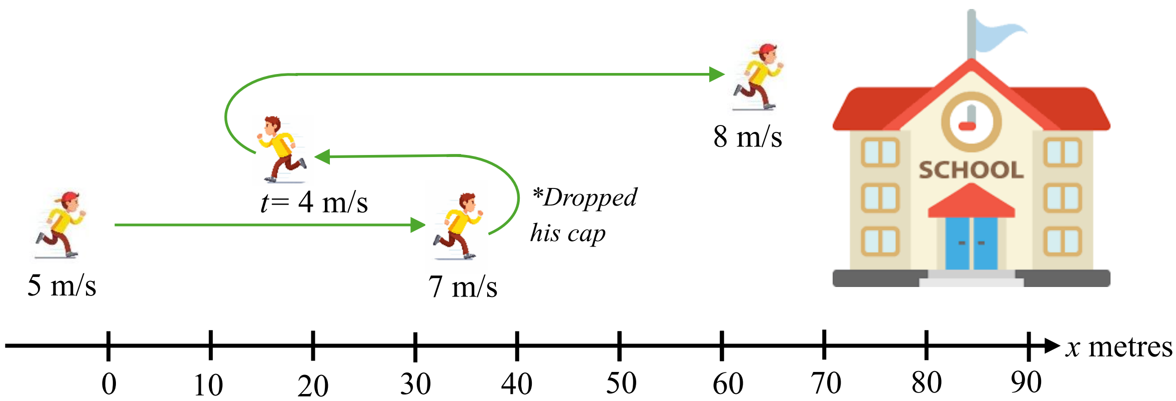
(B) Velocity and Acceleration

Instantaneous speed vs **instantaneous velocity** vs **average speed**

The diagram below shows a student running to school because he is late.



- The speed at a particular instant in time is called the i__________________ s_________. (e.g. 5 m/s, 4 m/s)
- Instantaneous speed with a direction is called the i__________________ v______________.

  e.g. 5 m/s (right), 4 m/s (left)

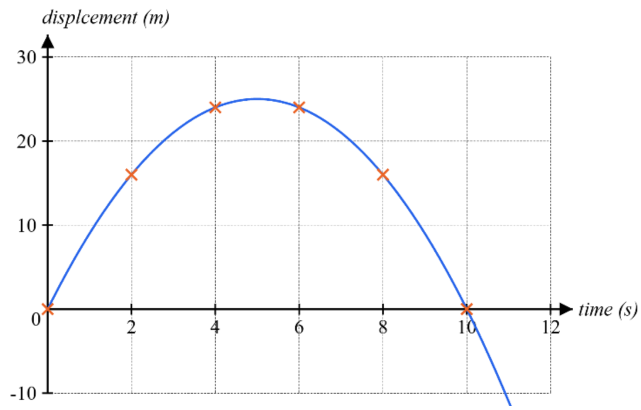
  or +5 m/s −4 m/s (positive means right while negative means left)

- The a______________ s_________ is the total distance divided by the total time taken to reach the school.

Note: in day-to-day life, when we say "what is the speed of the student?" we are referring to the average speed.

How can we find the instantaneous speed or velocity of the student?

Experimentally we can plot the displacement travelled every 5 seconds and model the running path by a curve:



In Secondary 3 Math, we learnt that the gradient of a distance-time graph is speed.

Similarly, the gradient of a **displacement-time** graph is **velocity**.

Hence, to find the instantaneous velocity at $t = 4$, we can draw a tangent line or apply what we learnt in differentiation.

$$\boxed{v = \frac{\mathrm{d}s}{\mathrm{d}t}}$$

- If $v < 0$, the body moves in the direction of negative displacement
- If $v = 0$ the body is **instantaneously at rest**
- if $v > 0$, the particle moves in the direction of positive displacement

Q) Is it true that if velocity is decreasing, the speed also decreases?

If v is a function of t, then the rate of change of velocity with respect to time is the acceleration.

$$\boxed{a = \frac{\mathrm{d}v}{\mathrm{d}t} =}$$

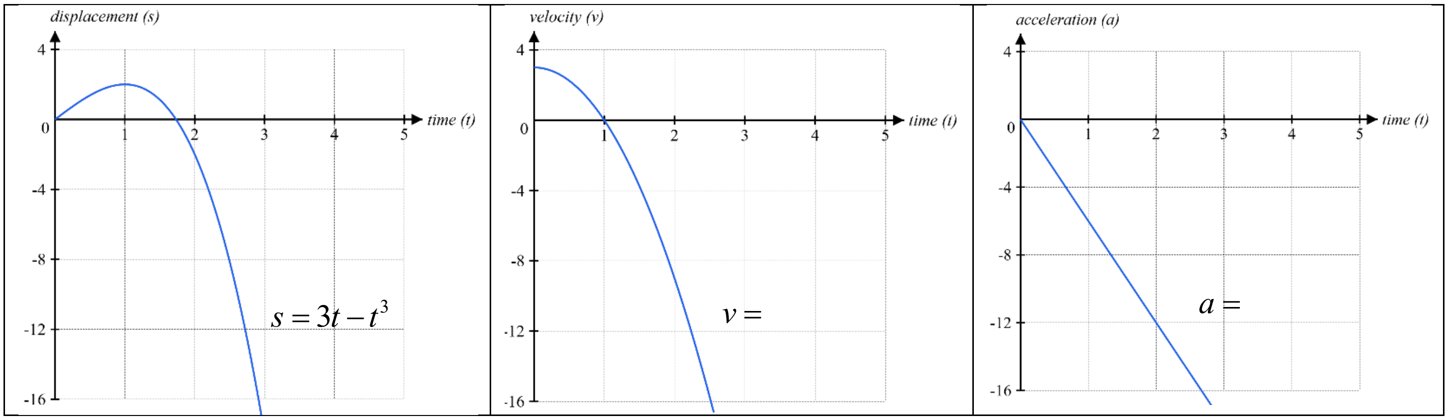
- if $a < 0$, the velocity decreases with respect to time
- if $a > 0$ the velocity increases with respect to time

Note: Acceleration is a vector quantity, not because of the direction of motion itself, but because it describes how velocity (a vector) is changing.

e.g. A boy slows down at a constant rate from 8 m/s at $t = 0$ to 2 m/s at $t = 3$. The acceleration is:

Summary:

$s = \mathrm{f}(t) \xrightarrow{v = \frac{ds}{dt}} v = \mathrm{f}(t) \xrightarrow{a = \frac{dv}{dt}} a = \mathrm{f}(t)$



$s = \mathrm{f}(t) \xleftarrow{s = \int v \, dt} v = \mathrm{f}(t) \xleftarrow{v = \int a \, dt} a = \mathrm{f}(t)$

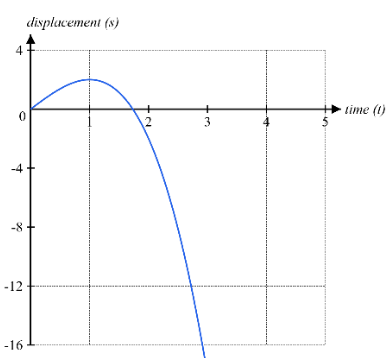
<u>What is **negative** displacement, **negative** velocity and **negative** acceleration?</u>

By convention, we take moving rightward as positive and moving upwards as positive

negative displacement	negative velocity	negative acceleration
The body is **located** in the negative *x* direction.	The body **moves** in the **direction of negative displacement** e.g. if rightward is positive, the body is moving towards the left e.g. if upwards is positive the body is moving downward	Velocity of object **decreases** with respect to time e.g. The displacement-time graph concave downwards

<u>Finding Distance Travelled:</u>

Given that the displacement of the object with respect to time is given by $s = 3t - t^3$, find the distance travelled by the object at $t = 2$.



16.2 Application of Differentiation in Kinematics

Finding velocity and acceleration given displacement

A particle *P* moves in a straight line so that, *t* seconds after passing a fixed point *O*, its displacement, *s* metres, is given by $s = t^3 - 6t^2 + 9t + 56$. Find

(i) the velocity of *P* when $t = 4$,

(ii) the acceleration of *P* when $t = 4$,

(iii) the values of *t* when *P* is at instantaneous rest,

(iv) the distance travelled by *P* in the first 2.5 seconds of its motion,

(v) the distance of *P* from *O* when $t = 2.5$,

(vi) the minimum velocity of the particle and show that it is a minimum.

EXAMPLE 1

A particle P moves in a straight line so that, t seconds after passing a fixed point O, its displacement, s metres, is given by $s = 4t^3 - 15t^2 + 18t$. Find

(i) the velocity of P when $t = 3$,

(ii) the acceleration of P when $t = 3$,

(iii) the values of t when P is at instantaneous rest,

(iv) the distance travelled by P in the first 2 seconds of its motion,

(v) the distance of P from O when $t = 2$.

Problems involving trigonometric expressions

A particle moves in a straight line so that, t seconds after leaving a fixed point A, its displacement, s metres, is given by $s = 6t + 2\cos 3t$. Find

(i) the initial position of the particle,

(ii) an expression, in terms of t, for the velocity and for the acceleration of the particle,

(iii) the distance of the particle form A when it first comes to rest.

EXAMPLE 2

A particle moves in a straight line so that, t seconds after leaving a fixed point B, its displacement, s metres, is given by $s = 3t + 4 \sin 2t$. Find

(i) the displacement of the particle when $t = 1.1$,

(ii) an expression, in terms of t, for the velocity and for the acceleration of the particle,

(iii) the distance of the particle from B when it first comes to rest.

Problems involving exponential expressions

A computer animation shows a cartoon rabbit moving in a straight line so that, t seconds after leaving a fixed point A, its displacement, s metres, is given by $s=6\left(\frac{2}{\mathrm{e}^{t}}-\frac{2}{\mathrm{e}^{2t}}\right)$. the rabbit comes to instantaneous rest at the point B to pick up a carrot.

(i) Show that the rabbit reaches B when $t=\ln 3$.

(ii) Find the average speed of the rabbit from A to B.

16.3 Application of Integration in Kinematics

We have learnt that integration is the reverse process of differentiation. Hence,

$$s = \int v \, \mathrm{d}t$$

$$v = \int a \, \mathrm{d}t$$

Finding displacement given velocity

A particle P moves in a straight line so that, t seconds after leaving a fixed point A, its velocity, v m s^{-1}, is given by $v = 2t^2 - 7t + 3$. When $t = 0$, its displacement from O is 5 metres. Find

(i) an expression, in terms of t, for the displacement of P from O,	**(ii)** the values of t when P is at instantaneous rest,
(iii) the distance of P from O at $t = 2$,	
(iv) the distance travelled by P in the first 5 seconds of its motion.	

EXAMPLE 3

A particle P moves in a straight line so that, t seconds after leaving a fixed point B, its velocity, v m/s, is given by $v = 4t^2 - 2t - 2$. When $t = 0$, its displacement from O is 9 metres. Find

(i) an expression, in terms of t, for the displacement of P from O,

(ii) the value of t when P is at instantaneous rest,

(iii) the distance of P from O at $t = 3$,

(iv) the distance travelled by P in the first 6 seconds of its motion.

Finding displacement and velocity given acceleration

A particle, moving in a straight line, passes a fixed point O with a velocity of 16 m/s. The acceleration, a m/s^2, of the particle, t s after passing through O, is given by $a = 2t - 10$. Find

(i) the values of t when the particle is at instantaneous rest,	**(ii)** the displacement of the particle when $t = 5$,
(iii) the distance travelled by the particle in the first 5 seconds of its motion,	
(iv) the distance travelled by the particle in the 5^{th} second.	

EXAMPLE 4

A particle, moving in a straight line, passes a fixed point O with a velocity of 30 cm/s. The acceleration, a cm/s^2, of the particle, t s after passing through O, is given by $a = 4t - 16$. Find

(i) the values of t when the particle is at instantaneous rest,

(ii) the displacement of the particle when $t = 3$,

(iii) the total distance travelled by the particle in the first 3 seconds of its motion,

(iv) the distance travelled by the particle in the 3rd second.

EXAMPLE 5

A particle travelling in a straight line passes through a fixed point O with a velocity of 2 m/s. The acceleration, a m/s^2, of the particle, t s after passing through O, is given by $a = -e^{-\frac{1}{3}t}$. The particle comes to instantaneous rest at the point P.

(i) Show that the particle reaches P when $t = 3 \ln 3$.

(ii) Calculate the distance OP.

(iii) Show that the particle is again at O at some instant during the ninth second after first passing through O.